

Problem-Solution Fit

Date	12 July 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience.

1. CUSTOMER SEGMENT(S) [CS] Self-paced online learners studying independently Teaching assistants (TAs) and academic support staff handling student queries	6. CUSTOMER LIMITATIONS (e.g., budget, devices) [CL] Only a personal laptop/browser available, no dedicated LMS integration Limited free-tier Gemini API quota constrains request volume No budget for large hosted backend infrastructure	5. AVAILABLE SOLUTIONS — PROS & CONS [AS] Manual TA/instructor replies — personal, but slow and inconsistent Generic chatbots/FAQ bots — instant, but emotion-blind and one-size-fits-all Peer forums — community knowledge, but unpredictable response time
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Students get stuck on concepts without quick, empathetic help — happens repeatedly through every course	9. PROBLEM ROOT / CAUSE [RC] No existing tool automatically senses a learner's emotional state from text, so both students and TAs must manually infer confusion or frustration, causing delay and inconsistency	7. BEHAVIOR + ITS INTENSITY [BE] Repeated re-reading of material (moderate intensity, 10–30 minutes before giving up) Delayed help-seeking — students wait until frustration is high before asking (high intensity)
3. TRIGGERS TO ACT [TR] Re-reading the same material without progress A forum question goes unanswered for hours A TA notices a growing backlog of unread student messages	10. YOUR SOLUTION [SL] The Emotion Detection & Learning Support Engine classifies a student's emotional state (Confused, Frustrated, Curious, Confident, Bored) from free text using BiLSTM + BERT, then instantly generates personalized, Gemini-powered guidance — removing the manual triage step for students and TAs alike, with every interaction logged and visualized for continuous improvement.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Course LMS, study forums, messaging TAs/instructors by chat or email
4. EMOTIONS BEFORE / AFTER [EM] Before: confused, frustrated, anxious, overwhelmed After: relieved, encouraged, confident, motivated to continue		OFFLINE In-person office hours, classroom discussions